

Acute Chest Syndrome



Section I: Scenario Demographics

Scenario Title:	Acute Chest Syndrome (Sickle Cell Crisis)		
Date of Development:	(DD/MM/YYYY)		
Target Learning Group:	☐ Juniors (PGY 1 - 2)	☒ Seniors (PGY $\geq$ 3)	☐ All Groups

Section II: Scenario Developers

Scenario Developer(s):	Carla Angleski
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Section III: Curriculum Integration

Learning Goals & Objectives	
Educational Goal:	To expose learners to an acute chest syndrome in sickle cell disease and review the management of acute chest syndrome.
CRM Objectives:	1. Allocate resources to ensure child is managed and mom is kept informed and calm. 2. Anticipate and appropriately prepare for the intubation of a child in sickle cell crisis
Medical Objectives:	1. Recognize acute chest syndrome and modify management accordingly 2. Perform safe intubation of a critically ill child

Case Summary: Brief Summary of Case Progression and Major Events
4-year-old boy with known sick cell disease presents with two days of cough and a one afternoon of fever. Patient is initially saturating at 88%, looks unwell and is in moderate-severe distress. During the case, the patient's oxygenation with drop and the emergency team is expected to provide airway support. They will also need to pick appropriate induction agents for intubation. The case will end with ICU admission. During the case, the mother will also be challenging/questioning the team until a team member is delegated to help keep the mother calm.

References
Howard et al. (2011). Guideline on the management of Acute Chest Syndrome in Sickle Cell Disease. British Society of Haematology
Marx, J. A., Hockberger, R. S., Walls, R. M., & Adams, J. (2013). <i>Rosen's emergency medicine: Concepts and clinical practice</i> . St. Louis: Mosby.

Section IV: Scenario Script



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A. Clinical Vignette: To Read Aloud at Beginning of Case

You are working the day shift at a tertiary children's hospital. A mother brings in her son, James, a four-year old boy with known sickle cell disease (HbSS). She is concerned since he's had low energy and a cough for two days. Now he's had a fever since this afternoon.

B. Scenario Cast & Realism

Patient:	☒ Pediatric Computerized Mannequin	Realism: <i>Select most important dimension(s)</i>	☒ Conceptual
	☐ Mannequin		☒ Physical
	☐ Standardized Patient		☐ Emotional/Experiential
	☐ Hybrid		☐ Other:
	☐ Task Trainer		☐ N/A

Confederates Brief Description of Role

Mother	Provide past medical history, behaves in challenging/questioning manner until a specific team member is assigned to keep mother informed and calm

C. Required Monitors

☒ EKG Leads/Wires	☐ Temperature Probe	☐ Central Venous Line
☒ NIBP Cuff	☐ Defibrillator Pads	☐ Capnography
☒ Pulse Oximeter	☐ Arterial Line	☐ Other:

D. Required Equipment

☒ Gloves	☒ Nasal Prongs	☐ Scalpel
☒ Stethoscope	☒ Venturi Mask	☐ Tube Thoracostomy Kit
☐ Defibrillator	☒ PEDS: Non-Rebreather Mask	☐ Cricothyroidotomy Kit
☒ IV Bags/Lines	☒ Bag Valve Mask	☐ Thoracotomy Kit
☒ IV Push Medications	☒ Laryngoscope	☐ Central Line Kit
☐ PO Tabs	☐ Video Assisted Laryngoscope	☐ Arterial Line Kit
☐ Blood Products	☒ ET Tubes	☐ Other:
☐ Intraosseous Set-up	☐ LMA	☐ Other:

E. Moulage

Darker skin mannequin wearing t-shirt and shorts.

F. Approximate Timing

Set-Up:	3 min	Scenario:	10 min	Debriefing:	15 min
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Section V: Patient Data and Baseline State

A. Patient Profile and History			
Patient Name: James Leroy		Age: 4	Weight: 22kg
Gender: ☒ M ☐ F		Code Status: Full Code	
Chief Complaint: Cough X 2 days and fever			
History of Presenting Illness: Patient was in usual state of health when he developed a non-productive cough 2 days ago, and a fever earlier today. He has generally been feeling unwell and unable to go to school. Patient is not his usual energetic self according to mom.			
Past Medical History:	Multiple previous admissions	Medications:	Prophylactic penicillin
	2 ICU admissions for chest crisis requiring exchange transfusion		Hydroxyurea (but not taking it consistently due to cost)
	Sepsis X1, Dactylitis X 1, Bony crisis X 5		
	Immunizations to date		
Allergies: nil			
Social History: Family immigrated from Zimbabwe two years ago. Lives at home with parents and 9-year-old sister. James is a student at elementary school.			
Family History: Both parents are identified as sickle cell carriers. Sister is a carrier.			
Review of Systems:	CNS:	Less energy than usual, but no other specific complaints.	
	HEENT:	Nasal congestion for 2 days	
	CVS:	Normal	
	RESP:	Non-productive cough for 2 days	
	GI:	Normal	
	GU:	Normal	
	MSK:	Normal	INT:
B. Baseline Simulator State and Physical Exam			
☐ No Monitor Display		☒ Monitor On, no data displayed	
☐ Monitor on Standard Display			
HR: 130 /min	BP: 95/25	RR: 30/min	O ₂ SAT: 88%
Rhythm: regular	T: 39.5 °C	Glucose: 7.2 mmol/L	GCS: 15 (E 4 V 5 M 6)
General Status: looks unwell, moderate-severe distress			
CNS:	Awake and alert		
HEENT:	Normal		
CVS:	Pulses bilaterally strong, cap refill <2 seconds		
RESP:	Diffuse crackles bilaterally with poor air entry to the right		
ABDO:	Soft, no distension, no pain on palpation		
GU:	Normal		
MSK:	No hot/swollen joints.	SKIN:	Pale

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Section VI: Scenario Progression

Scenario States, Modifiers and Triggers			
Patient State	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State	
1. Baseline State Rhythm: HR: 130/min BP: 95/25 RR: 30/min O ₂ SAT: 88% T: 39.5 °C	Looks unwell, mod-severe distress. Sitting in bed with mom at bedside. Difficulty responding to questions due to SOB.	<u>Learner Actions</u> - ☐ Full history relevant to sickle cell - ☐ Monitors on & apply oxygen - ☐ Airway assessment - ☐ Respiratory & CVS exam - ☐ Obtain IV access - ☐ Blood work (CBC with differential, gas, lytes, BUN, creat, gluc, retics, blood cultures, group and screen) - ☐ IV ABX (ceftriaxone & erythromycin) - ☐ IV fluid NS 10-20cc/kg bolus - ☐ CXR STAT	<u>Modifiers</u> <i>Changes to patient condition based on learner action</i> - If mother not addressed (or no team member assigned to her) → mother to become agitated & question management <u>Triggers</u> <i>For progression to next state</i> - 3 minutes → 2. Deterioration
2. Deterioration HR 140/min BP: 85/25 RR: 30/min O ₂ SAT: 84% by FM CVS: CRT 4 seconds	Poor oxygenation despite supplemental O ₂ , mom expressing concern about "the numbers being lower"	<u>Learner Actions</u> - ☐ Reassess airway: suction, reposition head, reapply O ₂ mask - ☐ Consider high flow O ₂ for CPAP or BVM with PEEP valve - ☐ Prepare for intubation - ☐ Explain possible intubation need to mom - ☐ Repeat NS bolus 10-20 cc/kg - ☐ Consider ABG	<u>Modifiers</u> - High flow or BVM with peep → O ₂ SAT to 90% - No intubation by 6 minutes → slowly drop O ₂ SAT to 80% <u>Triggers</u> - Start intubation → 3. Intubation
3. Intubation O ₂ SAT → as per end point of prior state	Same as previous state, sedated and paralyzed once RSI given	<u>Learner Actions</u> - ☐ Preparation/equipment - ☐ Intubates with appropriate induction and paralytic using ETT 5.0 or 5.5 cuffed (ketamine 1-2mg/kg = 22-44mg, midazolam 0.1mg/kg = 2.2mg, etomidate 0.3mg/kg = 6.6mg, succinylcholine 1.5mg/kg = 33mg, rocuronium 1mg/kg = 22mg) - ☐ Check tube placement with ETCO ₂ , auscultation and CXR	<u>Modifiers</u> - Drop O ₂ SAT by 4% during intubation <u>Triggers</u> - Intubation → 4. Post-Intubation
4. Post-intubation O ₂ SAT → 94% (increased over 90 seconds)	Give results of blood work at onset of state	- ☐ Call ICU if not already done - ☐ Inform mother of progress - ☐ Initiate sedation meds - ☐ Call for PRBC - ☐ Call for exchange transfusion	END CASE PRN

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Section VII: Supporting Documents, Laboratory Results, & Multimedia

Laboratory Results						
Na: 144	K: 5	Cl: 102	HCO ₃ : 16	BUN: 7	Cr: 70	Glu: 6.9
ABG	pH: 7.2	PCO ₂ : 55	PO ₂ : 84	HCO ₃ : 16		
WBC: 8		Hg: 62		Plt: 165		

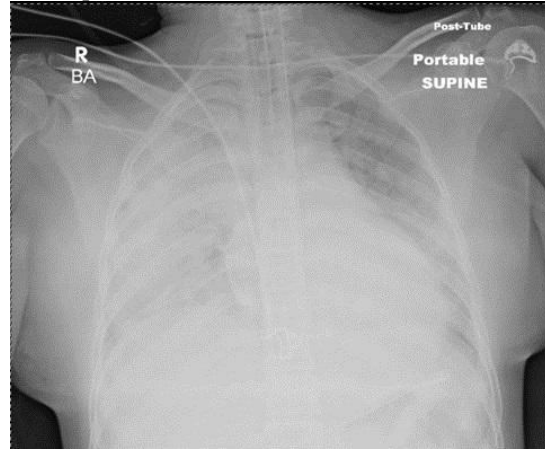
Images (ECGs, CXRs, etc.)

CXR: R-sided infiltrate



<http://reference.medscape.com/features/slideshow/sickle-cell#8>

CXR: post-intubation



<http://www.swjpcc.com/critical-care/?currentPage=4>

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Section VIII: Debriefing Guide

General Debriefing Plan	
☐ Individual	☒ Group
☐ With Video	☒ Without Video
Objectives	
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Sample Questions for Debriefing	
1) What criteria define an acute chest syndrome? 2) When should transfusion be considered in acute chest syndrome? 3) How much fluid was given to the child during the case? What is the role for fluid in acute chest syndrome? 4) What strategies do you have for addressing parents of critically ill children? 5) Do you feel your team communicated well? Did all team members know what was being treated? 6) Did everyone feel like they had a clearly defined role? What strategies could be used in the future to ensure roles are clear?	
Key Moments	
Identification of acute chest syndrome	
Identifying need to intubate and safe peri-intubation management	
Addressing mom in a calm manner	